PRELIMINARY STATISTICAL COMPUTATION

ASD Risk Score Cutoff Development for AuRA

1. Purpose of the Computation

The purpose of this preliminary computation is to determine possible score cutoff points for classifying toddler behavioral questionnaire scores into three preliminary risk levels: Low, Moderate, and High. The proposed procedure is descriptive analysis followed by two binary ROC analyses and Youden's Index. The resulting cutoffs are preliminary and will be presented to the Mathematics/Statistics professor for validation.

2. Data Used

Group	Number of Respondents	Percentage
No Concern	193	38.60%
Not Evaluated	251	50.20%
Diagnosed ASD	56	11.20%
TOTAL	500	100.00%

Check:

193+251+56=500

Percentage Formula:

Percentage= Frequency/Total x 100

No Concern

193/500×100=38.6%

Not Evaluated

251/500×100=50.2%

Diagnosed ASD

56/500×100=11.2%

Check:

38.6 + 50.2 + 11.2 = 100%

3. Scoring System

The questionnaire has 13 items with a total possible score of 0 to 26.

Items	Response	Score
Q1–Q9	Always / Palagi	0
Q1–Q9	Sometimes / Minsan	1
Q1–Q9	Never / Hindi	2
Q10–Q13	No / Wala	0
Q10–Q13	Sometimes / Minsan	1
Q10–Q13	Yes / Meron	2

Total Score
Total Score = Q1 + Q2 + Q3 + … + Q13

Therefore: Minimum score = 0; Maximum score = 26.

4. Example of Total Score Computation

Question	Response	Score
Q1	Always	0
Q2	Sometimes	1
Q3	Always	0
Q4	Never	2
Q5	Always	0
Q6	Sometimes	1
Q7	Never	2
Q8	Always	0
Q9	Always	0
Q10	No	0
Q11	Sometimes	1
Q12	Yes	2
Q13	No	0
TOTAL		9

Example calculation: 0 + 1 + 0 + 2 + 0 + 1 + 2 + 0 + 0 + 0 + 1 + 2 + 0 = 9.

5. Descriptive Statistics

The following statistics were computed for each existing group: mean, median, mode, standard deviation, minimum, and maximum.

Group	n	Mean	Median	Mode	SD	Min	Max
No Concern	193	6.82	6	4	4.34	0	21
Not Evaluated	251	8.62	8	5, 9, 11	4.58	0	24
Diagnosed ASD	56	14.54	14	19	6.26	4	26

6. Sample Descriptive Computations

Mean
Mean = Σx / n

No Concern: 1317 / 193 = 6.8238 ≈ 6.82

Not Evaluated: 2163 / 251 = 8.6175 ≈ 8.62

Diagnosed ASD: 814 / 56 = 14.5357 ≈ 14.54

Median
Odd n: Median position = (n + 1) / 2
Even n: Median = (middle score 1 + middle score 2) / 2

No Concern: (193 + 1) / 2 = 97th score → Median = 6

Not Evaluated: (251 + 1) / 2 = 126th score → Median = 8

Diagnosed ASD: (28th + 29th scores) / 2 = (14 + 14) / 2 = 14

Standard Deviation
$s = \sqrt{[\ \Sigma(x - \bar{x})^2 / (n - 1)]}$

Computed results:
No Concern = 4.34;
Not Evaluated = 4.58;
Diagnosed ASD = 6.26.

7. Why ROC Analysis?

Because the study has three existing groups, the proposed method uses two binary comparisons to obtain two cutoff points:

1. Cutoff 1: No Concern vs Not Evaluated

2. Cutoff 2: Not Evaluated vs Diagnosed ASD

For each cutoff c: score ≥ c is treated as Positive, while score < c is treated as Negative.

8. Cutoff 1: No Concern vs Not Evaluated

At cutoff = 9:

	Predicted Negative	Predicted Positive
No Concern	TN = 142	FP = 51
Not Evaluated	FN = 133	TP = 118

Sensitivity
TP / (TP + FN) = 118 / (118 + 133) = 118 / 251 = 0.4701 = 47.01%

Specificity
TN / (TN + FP) = 142 / (142 + 51) = 142 / 193 = 0.7358 = 73.58%

Youden's Index
J = Sensitivity + Specificity − 1
J = 0.4701 + 0.7358 − 1 = 0.2059

The highest Youden's Index in this first comparison occurs at cutoff 9. Therefore, proposed Cutoff 1 = 9.

For Cutoff 1:

Cutoff	Sensitivity	Specificity	Youden J
5	80.48%	34.72%	0.1519
6	71.31%	44.56%	0.1587
7	62.95%	52.33%	0.1528
8	54.98%	64.25%	0.1923
9	**47.01%**	**73.58%**	**0.2059**
10	37.85%	77.20%	0.1505
11	32.27%	81.35%	0.1362
12	23.11%	84.46%	0.0756
13	19.12%	88.60%	0.0772

Cutoff	Sensitivity	Specificity	Youden J
14	15.54%	91.19%	0.0673

9. Cutoff 2: Not Evaluated vs Diagnosed ASD

At cutoff = 12:

	Predicted Negative	Predicted Positive
Not Evaluated	TN = 193	FP = 58
Diagnosed ASD	FN = 20	TP = 36

Sensitivity
TP / (TP + FN) = 36 / (36 + 20) = 36 / 56 = 0.6429 = 64.29%

Specificity
TN / (TN + FP) = 193 / (193 + 58) = 193 / 251 = 0.7689 = 76.89%

Youden's Index
J = Sensitivity + Specificity − 1
J = 0.6429 + 0.7689 − 1 = 0.4118

The highest Youden's Index in this second comparison occurs at cutoff 12. Therefore, proposed Cutoff 2 = 12.

Cutoff	Sensitivity	Specificity	Youden J
5	96.43%	19.52%	0.1595
6	94.64%	28.69%	0.2333
7	91.07%	37.05%	0.2812
8	85.71%	45.02%	0.3073
9	76.79%	52.99%	0.2977
10	71.43%	62.15%	0.3358
11	71.43%	67.73%	0.3916
12	**64.29%**	**76.89%**	**0.4118**
13	57.14%	80.88%	0.3802
14	51.79%	84.46%	0.3625
15	48.21%	87.65%	0.3586
16	44.64%	90.84%	0.3548
17	39.29%	94.02%	0.3331
18	33.93%	96.02%	0.2994

10. Comparison of Candidate Cutoffs

Comparison	Best Cutoff	Sensitivity	Specificity	Youden J
No Concern vs Not Evaluated	9	47.01%	73.58%	0.2059
Not Evaluated vs Diagnosed ASD	12	64.29%	76.89%	0.4118

11. Proposed Preliminary Risk Ranges

Total Score	Preliminary Risk Level
0–8	Low
9–11	Moderate
12–26	High

These ranges are preliminary score-based categories derived from the proposed ROC/Youden procedure. They are not presented as clinical diagnostic cutoffs.

12. Final Preliminary Result

Item	Result
Total Respondents	500
No Concern	193
Not Evaluated	251
Diagnosed ASD	56
Proposed Cutoff 1	9
Proposed Cutoff 2	12
Preliminary Low Range	0–8
Preliminary Moderate Range	9–11
Preliminary High Range	12–26